

NOVOSIBIRSK OGOURTSOVO, Russian Federation

WMO: 296380

Lat: 54.9108N

Lon: 82.9569E

Elev: 131

StdP: 99.76

Time Zone: 7.00 (E07)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.9	(c) -33.1	(d) -38.9	(e) 0.1	(f) -35.8	(g) -35.9	(h) 0.1	(i) -33.0	(j) 9.1	(k) -6.4	(l) 8.2	(m) -9.0	(n) 1.1	(o) 210	(p) 0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 29.1	(d) 18.9	(e) 27.5	(f) 18.4	(g) 26.0	(h) 17.8	(i) 20.6	(j) 26.2	(k) 19.7	(l) 25.0	(m) 18.9	(n) 24.1	(o) 2.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.8	22.8	17.8	13.0	22.2	16.9	12.3	21.3	60.2	26.0	57.2	25.1	54.5	24.2	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 7.1	(c) 6.3	(e) -37.3	(f) 31.7	(g) 3.4	(h) 1.7	(i) -39.8	(j) 32.8	(k) -41.8	(l) 33.8	(m) -43.7	(n) 34.7	(o) -46.2	(p) 35.9		
(a) 8.1	(b) 7.1	(c) 6.3	(e) -37.4	(f) 22.2	(g) 3.4	(h) 1.1	(i) -39.9	(j) 23.0	(k) -41.9	(l) 23.6	(m) -43.8	(n) 24.2	(o) -46.3	(p) 25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-18.5	-16.3	-5.9	4.2	10.7	17.7	19.3	16.6	10.7	3.6	-6.2	-13.9
	DBStd	14.37	8.65	7.43	5.71	5.74	5.02	4.28	2.95	3.49	4.51	4.95	7.25	8.48
	HDD10.0	3836	883	736	494	188	54	3	0	1	44	205	486	742
	HDD18.3	6112	1141	969	752	425	240	61	25	76	231	456	736	1000
	CDD10.0	892	0	0	0	13	76	235	289	206	67	7	0	0
	CDD18.3	126	0	0	0	0	3	43	56	22	3	0	0	0
	CDH23.3	1101	0	0	0	4	53	402	413	191	38	0	0	0
	CDH26.7	226	0	0	0	0	8	88	84	37	9	0	0	0
Wind	WSAvg	2.9	2.9	2.8	3.1	3.1	3.3	2.4	2.1	2.3	2.5	3.0	3.4	3.3
Precipitation	PrecAvg	612	41	29	37	37	49	63	78	56	46	50	69	58
	PrecMax	756	69	58	81	93	111	127	140	99	104	124	136	117
	PrecMin	456	9	2	5	6	14	27	26	15	17	24	27	24
	PrecStd	83	17	13	19	18	25	24	28	20	19	21	25	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.0	7.9	23.2	27.6	30.6	31.2	30.3	27.8	19.4	8.5	1.5
		MCWB	-0.2	-0.5	3.7	12.7	14.4	19.2	19.9	19.6	17.1	11.8	6.3	0.5
	2%	DB	-1.7	-0.8	4.9	19.0	24.6	29.0	28.7	27.4	23.4	16.2	6.3	0.4
		MCWB	-2.6	-1.9	1.6	11.0	13.4	18.4	19.5	18.6	14.9	10.0	4.2	-0.3
	5%	DB	-3.6	-2.6	3.4	16.2	22.5	27.5	27.2	25.5	20.6	13.5	3.8	-0.7
		MCWB	-4.4	-3.5	1.1	9.6	12.9	18.1	19.0	17.9	13.7	8.8	2.4	-1.4
	10%	DB	-6.0	-5.1	2.0	12.9	19.7	25.8	25.7	23.5	18.2	10.7	2.0	-2.7
		MCWB	-6.8	-6.1	0.3	8.0	11.7	17.4	18.6	16.9	12.7	7.4	0.8	-3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	-0.2	3.8	13.6	17.0	21.2	22.1	21.4	17.7	12.4	6.7	0.7
		MCDB	1.0	0.6	7.2	20.8	24.2	28.2	27.4	27.0	24.9	17.8	8.1	1.1
	2%	WB	-2.6	-1.9	2.2	11.8	14.8	19.9	20.9	19.7	15.7	10.6	4.4	-0.1
		MCDB	-1.8	-0.8	4.1	18.1	22.2	26.3	26.3	25.0	21.4	15.2	5.9	0.4
	5%	WB	-4.4	-3.6	1.2	10.0	13.7	19.0	20.1	18.8	14.5	9.3	2.5	-1.4
		MCDB	-3.7	-2.6	3.1	15.3	20.5	25.3	25.3	24.1	19.8	13.0	3.9	-0.7
	10%	WB	-6.7	-6.1	0.3	8.1	12.3	18.1	19.4	17.8	13.2	7.6	0.9	-3.5
		MCDB	-6.0	-5.2	1.9	12.9	19.1	24.1	24.0	22.5	17.5	10.5	1.8	-2.7
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.0	9.1	9.7	10.6	10.2	9.5	9.9	9.6	7.1	5.7	7.3
		MCDBR	7.8	7.8	9.4	14.6	14.7	13.1	12.0	12.9	14.1	12.5	5.6	5.8
		MCWBR	7.6	7.5	7.1	8.1	6.7	5.2	4.6	5.5	6.6	6.9	4.4	5.8
	5% WB	MCDBR	8.0	8.0	7.7	13.6	12.4	11.2	10.0	11.3	12.1	11.3	5.6	5.9
		MCWBR	7.8	7.7	6.3	7.9	6.8	4.9	4.5	5.2	6.0	6.6	4.5	6.0
Clear-Sky Solar Irradiance	taub	0.294	0.329	0.378	0.409	0.407	0.410	0.437	0.413	0.373	0.355	0.328	0.297	
	taud	2.004	1.958	1.907	2.080	2.211	2.287	2.210	2.262	2.318	2.231	2.040	1.954	
	Ebn at Noon	589	707	761	806	836	837	802	799	782	689	537	475	
	Edh at Noon	72	109	145	141	131	123	131	117	97	83	70	61	
All-Sky Solar Radiation	RadAvg	0.95	1.98	3.45	4.65	5.35	6.03	5.40	4.56	3.08	1.54	0.76	0.61	
	RadStd	0.08	0.11	0.26	0.29	0.59	0.53	0.35	0.34	0.36	0.21	0.13	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page